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The severity of psychiatric disorders

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Abstract

The issue of the severity of psychiatric disorders has great clinical importance. For example, severity influences decisions about level of care, and affects decisions to seek government assistance due to psychiatric disability. Controversy exists as to the efficacy of antidepressants across the spectrum of depression severity, and whether patients with severe depression should be preferentially treated with medication rather than psychotherapy. Measures of severity are used to evaluate outcome in treatment studies and may be used as meaningful endpoints in clinical practice. But, what does it mean to say that someone has a severe illness? Does severity refer to the number of symptoms a patient is experiencing? To the intensity of the symptoms? To symptom frequency or persistence? To the impact of symptoms on functioning or on quality of life? To the likelihood of the illness resulting in permanent disability or death? Putting aside the issue of how severity should be operationalized, another consideration is whether severity

should be conceptualized similarly for all illnesses or be disorder specific. In this paper, we examine how severity is characterized in research and contemporary psychiatric diagnostic systems, with a special focus on depression and personality disorders. Our review shows that the DSM-5 has defined the severity of various disorders in different ways, and that researchers have adopted a myriad of ways of defining severity for both depression and personality disorders, although the severity of the former was predominantly defined according to scores on symptom rating scales, whereas the severity of the latter was often linked with impairments in functioning. Because the functional impact of symptom-defined disorders depends on factors extrinsic to those disorders, such as self-efficacy, resilience, coping ability, social support, cultural and social expectations, as well as the responsibilities related to one's primary role function and the availability of others to assume those responsibilities, we argue that the severity of such disorders should be defined independently from functional impairment.

Keywords: Severity, psychiatric disorders, functional impairment, symptoms, depression, personality disorders, transdiagnostic models, HiTOP, DSM-5, ICD-10

The determination of illness severity has important clinical implications. Depending on the disorder, severity affects decisions to seek treatment, the type and intensity of treatment, and whether to continue or stop treatment. Severity also impacts expectations in the fulfillment of role function and disability status. Measures of severity are used to evaluate outcome in treatment studies and may be used as meaningful endpoints in clinical practice.

But, what does it mean to say that someone has a severe illness? Of the various dictionary definitions of “severe”, the one that is most relevant to the characterization of illness is “of great degree”. This definition, however, does not convey what is meant when an illness is considered “severe”. Does severity refer to the number of symptoms a patient is experiencing? To the intensity of the symptoms? To symptom frequency or persistence? To the impact of symptoms on functioning or quality of life? To the likelihood of the illness resulting in permanent disability or death?

Some of these questions about the meaning of severity can be further elaborated. For

example, with regards to the prediction of mortality, does severity allude to imminent death, death in the near future, or death at any time in the future? Also, should the impact of intervention be considered? That is, is an illness severe only when death is likely if the illness is left untreated, or only if death is likely regardless of intervention?

Perhaps severity determinations should be independent of functional impact or prognosis and instead should be based on structural or morphological changes and damage to the diseased organ. To be sure, this is not relevant for many illnesses, but, when it can be measured, should this be the guiding principle for rating illness severity?

Putting aside the issue of how severity should be operationalized, another consideration is whether severity should be conceptualized similarly for all illnesses or be disorder specific. Should the severity of heart failure, rheumatoid arthritis, diabetes, an acute upper respiratory tract infection, and a headache be judged according to a common standard or metric, or should each disorder have its own respective guidelines for rating severity?

In this paper, we examine how severity is characterized in psychiatric research and contemporary psychiatric diagnostic systems. To illustrate some of the issues and controversies in determining the severity of psychiatric disorders, we focus on depression and personality disorders (PDs). The clinical significance of considering the severity of depression is reflected in official treatment guidelines wherein recommendations are based on illness severity^{1, 2}. The importance of considering the severity of PDs is reflected by the ICD-11 proposal to replace the specified criteria for different disorders by a single personality disorder category that is graded according to levels of severity^{3, 4}.

Before discussing the issue of severity of psychiatric disorders, we present a brief overview of how severity has been conceptualized, assessed and measured for various physical illnesses, highlighting the variability of approaches.

SEVERITY OF PHYSICAL ILLNESSES

There is no consensus or uniform overriding principle in distinguishing between levels of severity of physical illnesses. In some cases, severity is defined by the degree of structural

damage to the diseased organ. For example, the severity of rheumatoid arthritis has been defined according to radiographic evidence of joint damage⁵. The severity of diabetic retinopathy has been graded according to the degree of retinal damage assessed in a direct clinical eye exam⁶. In a related manner, physiological measures representing the impact of disease on the organ have been used to characterize the severity of some diseases. For example, left ventricular ejection fraction has been used as an index of the severity of cardiovascular disease^{7, 8, 9, 10}. Forced expiratory volume has been used as index of severity of cystic fibrosis¹¹. Aminotransferase and bilirubin levels have been used to assess the severity of hepatitis¹².

Sometimes severity is defined by a disorder-specific clinical examination. For example, not only have radiographic assessments been used to evaluate the severity of rheumatoid arthritis, but severity has additionally been defined according to a count of the number of swollen and painful joints¹³.

Illness severity has also been defined more broadly to encompass indices of the diseased organ as well as related and downstream effects. In a study of the prognostic implications of post-cardiac arrest illness severity, severity scores were based on cardiopulmonary dysfunction and neurologic status^{14, 15}. The severity of sickle cell disease has been based on the presence and frequency of complications such as renal failure, necrosis of hips and shoulders, and gallstones¹⁶. In studies of the severity of chronic obstructive pulmonary disease, the BODE index (B, body mass index; O, obstruction of airways as measured by forced expiratory volume in one second; D, dyspnea scale; E, exercise capacity as measured by a six-minute walk test) includes and goes beyond a direct, specific, assessment of pulmonary damage and has been found to be a better predictor of mortality, hospitalization, quality of life, and depression than forced expiratory volume alone¹⁷. The Unified Parkinson's Disease Rating Scale contains four subscales assessing mental state, activities of daily living, motor examination, and complications^{18, 19}.

Moving further away from a direct or physiological assessment of the diseased organ, the New York Heart Association Functional Classification is a measure of cardiac disease severity based on limitations in physical activities and the presence of physical symptoms associated with varying degrees of activity²⁰.

In contrast to disorder-specific physical and physiological indicators of severity, there are composite measures of overall illness severity, such as the Acute Physiology and Chronic Health Evaluation (APACHE) scores and the Simplified Acute Physiology Score (SAPS), based on non-specific clinical and biological indicators of health status such as body temperature, age, history of organ failure, electrolytes, and hematocrit^{21, 22}. These illness severity measures have been used to predict mortality in heterogeneous and single disorder samples of acutely ill emergency department and hospitalized patients^{23, 24}.

Finally, self-report questionnaires have been developed to assess the severity of some physical illnesses. The severity of benign prostatic hypertrophy as assessed by the American Urological Association Symptom Index is based on the frequency of symptoms²⁵. The Tinnitus Severity Index is based on the frequency of functional impairment or psychological symptoms due to tinnitus²⁶. The Bowel Symptom Severity Scale assesses the frequency, distress and disability of symptoms associated with irritable bowel syndrome²⁷. The severity of headaches as measured by the Headache Impact Questionnaire is a composite measure of headache frequency, the average pain intensity of headaches, and the impairment resulting from headaches²⁸. The Liverpool Seizure Severity Scale assesses perceptions of seizure control and severity of ictal and postictal symptoms²⁹.

Clark et al³⁰ summarized the approach taken to develop self-report measures of illness severity for six disease states studied in the Veterans Health Study. They defined illness severity in terms of patients' perceptions of the magnitude of symptoms or complications of the illness that are associated with reductions in health-related quality of life or health status. They distinguished disease severity from the impact of disease (e.g., impairment, life satisfaction, well-being), because the impact of disease is often mediated by personal characteristics (e.g., resiliency, self-efficacy) and social context.

SEVERITY OF PSYCHIATRIC DISORDERS AS DESCRIBED IN DSM-5

In contrast to some physical illnesses, there are no specific or non-specific biomarkers of psychiatric disorders that validly characterize the severity of the disorder. In the absence of such biological or structural indicators, researchers and clinicians are left to assess the epiphenomena of a psychiatric disorder to judge its severity.

Discussions of resource allocation in the public health sector often focus on patients with severe mental illness, though there is no consensus in how to define such an illness^{31, 32}. The DSM-5³³, like its immediate predecessors, defines severity for only some disorders. Table 1 lists the DSM-5 disorders with defined levels of severity.

Table 1.

Characterization of disorder severity in DSM-5

DSM-5 disorder	Features used to define severity
Major depressive disorder	Number of symptoms, level of distress caused by intensity of symptoms, and impairment in social and occupational functioning
Mania, hypomania	Same as major depressive disorder
Alcohol use disorder	Number of criteria
Drug use disorder	Number of criteria
Bulimia nervosa	Frequency of compensatory behaviors per week
Anorexia nervosa	Body mass index
Binge eating disorder	Frequency of eating binges
Learning disorders	Severity of deficit in learning skills and likelihood of learning the skills with or without intervention
Attention-deficit/hyperactivity disorder	Number of symptoms, severity of individual symptoms, or level of impairment caused by the symptoms
Intellectual disability	Level of adaptive functioning
Autism spectrum disorder	Degree of impairment in functioning due to deficits in verbal and nonverbal communication, inflexibility of behavior, difficulty coping with change, or restricted/repetitive behaviors

Stereotypic movement disorder	The ease by which the symptoms can be suppressed and the need for intervention to prevent serious injury
Psychotic disorders	Quantitative assessment on 5-point scale of primary feature of the psychosis (delusions, hallucinations, disorganized speech, abnormal psychomotor behavior, and negative symptoms). Rating is based on symptom intensity or subjective distress due to symptom
Reactive attachment disorder	Only the severe type is defined. Severe is defined as all criteria met at a high level
Disinhibited social engagement disorder	Only the severe type is defined. Severe is defined as all criteria met at a high level
Somatic symptom disorder	Number of criteria and somatic complaints
Psychological factors affecting other medical conditions	Degree of impact on medical condition or medical risk
Hypersomnolence disorder	Number of days per week with difficulty maintaining daytime alertness
Narcolepsy	Frequency of cataplexy and responsiveness of cataplexy to medication, number of naps per day, degree of disturbance of nocturnal sleep
Obstructive sleep apnea/hypopnea	Apnea/hypopnea index score
Nightmare disorder	Frequency of nightmares per week
Sexual disorders	Degree of distress related to symptoms
Premature ejaculation	Time to ejaculation
Substance/medication-induced sexual dysfunction	Percentage of occasions of sexual activity that dysfunction occurs
Oppositional defiant disorder	Number of settings in which the symptoms occur
Conduct disorder	Number of conduct problems or the degree of

Neurocognitive disorders	harm caused to others Degree of difficulty with instrumental activities of daily living
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The DSM-5 approach towards defining severity varies across disorders. The four severity levels of intellectual disability (mild, moderate, severe, profound) are the most elaborately defined, with three pages of descriptions of the adaptive functioning deficits characteristic of each level of severity. DSM-5 notes that severity was defined according to adaptive functioning level rather than IQ scores because the former is a better determinant of the level of supports that are needed. Similarly, the level of deficits and functional impairment defining the severity of autism spectrum disorders is linked to the supports required. The severity of learning disorders refers to the difficulties in learning skills as well as the likelihood of learning those skills with or without intervention. For example, DSM-5 defines severe impairment of a learning disorder as “severe difficulties learning skills, affecting several academic domains, so that the individual is unlikely to learn those skills without ongoing intensive individualized and specialized teaching for most of the school years”. For these disorders, then, the severity specifier is explicitly linked to suggested levels of intervention.

Depression and mania are classified as mild, moderate or severe according to the number of symptoms, the level of distress caused by the intensity of the symptoms, and the degree of impairment in social and occupational functioning. The severity of alcohol and drug use disorders is based on the number of criteria that are met (mild: 2 or 3 criteria; moderate: 4 or 5 criteria; severe: 6 or more criteria). The severity of attention-deficit/hyperactivity disorder is based on the number of symptoms, severity of individual symptoms, or level of impairment caused by the symptoms. The severity of bulimia nervosa is operationalized according to the number of inappropriate compensatory behaviors per week (mild: 1-3; moderate: 4-7; severe: 8-13; extreme: 14 or more), though the severity designation could be increased to reflect other symptoms or level of functional impairment. For anorexia nervosa, severity is defined according to body mass index, and for binge eating disorder it

is defined by the number of binge eating episodes per week, though, similar to bulimia nervosa, the severity designation can be increased to reflect other symptoms or degree of functional impairment. Severity of sexual disorders is based on the level of distress regarding the symptoms, except for premature ejaculation, for which severity is based on the time to ejaculation. The severity of cataplexy is based, in part, on lack of responsiveness to medication.

This brief overview illustrates the variability in the approaches taken in the DSM-5 towards defining degrees of severity, with some definitions emphasizing the number of criteria met, some others emphasizing the core feature of the disorder, some based on level of distress, and some focusing on response to intervention and prediction of course. In contrast to many physical illnesses, none of the definitions of severity refer to the likelihood of imminent or distal mortality, and most definitions do not refer to prognosis or future course. Rather, most definitions of severity in DSM-5 refer to the number of symptoms or criteria of the disorder, the frequency of symptoms, and the level of impairment or distress.

SEVERITY OF DEPRESSION

We focus on the severity of depression because it has received the most extensive research. While the research has not been entirely consistent, the severity of depression has been associated with health-related quality of life³⁴, functional impairment^{35, 36}, suicidality^{37, 38, 39}, longitudinal course^{40, 41, 42, 43}, and several biological variables^{44, 45, 46}. Moreover, the severity of depression has been at the core of controversies regarding the efficacy of treatment and whether certain forms of treatment should be recommended as first line interventions. Almost all research on severity is based on scores on depression symptom scales, though most scales have been developed without consideration as to how to best conceptualize and assess the severity of depression.

Severity levels of depression in DSM-5 and ICD-10

Three elements are used to define the severity levels of depression in DSM-5: the number of symptoms, the level of distress caused by the intensity of the symptoms, and the degree

of impairment in social and occupational functioning. The severity categorization applies to all depressive disorders, not just major depressive disorder (MDD). Mild depression is specified when “few, if any, symptoms in excess of those required to make the diagnosis are present, the intensity of the symptoms is distressing but manageable, and the symptoms result in minor impairment in social or occupational functioning”. Severe depression is specified when “the number of symptoms is substantially in excess of that required to make the diagnosis, the intensity of the symptoms is seriously distressing and unmanageable, and the symptoms markedly interfere with social and occupational functioning”. The DSM-5 does not explicitly define moderate depression other than to say that the number of symptoms, intensity of symptoms, and/or functional impairment are between mild and severe.

There are some problems with the DSM-5 specification of severity levels. The same definition of the severity specifier is used for MDD and persistent depressive disorder. This is a problem, because persistent depressive disorder requires fewer symptoms than does MDD to meet the DSM-5 diagnostic threshold. Thus, a patient with persistent depressive disorder who experiences the same number of symptoms as a patient with MDD, and with similar levels of functional impairment and distress, may be classified as more severe because the symptom count may be “substantially in excess” of the diagnostic threshold for persistent depressive disorder but not for MDD.

Another problem with the DSM-5 severity specifier is that the definition of functional impairment is limited to social or occupational functioning. This is inconsistent with the wording of the impairment criterion for the diagnosis of MDD and persistent depressive disorder, which refers to impairment in social, occupational, or *other important areas of functioning*. Thus, individuals who maintain social contacts, are not expected to be employed, but are unable to function as students or full-time parents, could be misclassified as less severe than they actually are.

While moderate severity is not specifically defined, the internal logic of the wording of the moderate severity description has a minor flaw. Mild depression requires low levels of symptoms, distress *and* functional impairment. Conversely, severe depression requires high levels of all three. Thus, moderate depression should be defined as lying between the mild and severe levels in symptoms, distress *or* functional impairment (not *and/or* as

DSM-5 defines it).

Finally, two other variables often considered important in discussions about depression severity – suicidality and need for hospitalization – are not considered in DSM-5's definition of severity.

What evidence supports the validity of the DSM-5 approach towards defining severity in this manner? One study from a population-based registry of twins who experienced a major depressive episode in the year prior to the interview found that the three aspects of the severity specifier – number of symptoms, severity of symptoms, and degree of functional impairment – were significantly, albeit only modestly, correlated⁴⁷. The authors concluded that the DSM severity construct was multifaceted and heterogeneous.

A study of psychiatric outpatients with a mood disorder⁴⁸, 84% of whom were in a major depressive episode, found that the number of DSM-IV symptoms of MDD was weakly correlated with clinicians' ratings on the Clinical Global Impression (CGI)⁴⁹ and the Global Assessment of Functioning (GAF)⁵⁰. Moreover, the severity ratings of some individual symptoms of depression were as highly correlated with CGI and GAF scores as was the total number of depressive symptoms. A small study of psychiatric inpatients with MDD found that the number of MDD criteria was weakly correlated with the Global Assessment Scale⁵¹. Kessler et al⁵² analyzed data from the National Comorbidity Study (NCS) and found that, compared to individuals who reported five or six MDD criteria during their worst episode of depression, individuals who reported seven to nine MDD criteria experienced more psychosocial impairment, more episodes of depression, and greater chronicity. Wakefield and Schmitz^{53, 54} examined the NCS database as well as another epidemiological survey and suggested that the number of depressive symptoms was less important than the type of depressive symptoms and other features of complicated depression in predicting future occurrence of a major depressive episode, seeking professional help for depression, a history of suicide attempt, and a history of psychiatric hospitalization. Thus, symptom count does not seem to be an adequate indicator of depression severity.

The ICD-10⁵⁵ designates three levels of severity – mild, moderate and severe – based on number of symptoms, severity of symptoms, functional impairment, level of distress and,

indirectly, type of symptoms. In contrast to DSM-5, there is no symmetry in the descriptions of the three levels of severity. Mild depression refers to the presence of two or three symptoms that are distressing though the patient is likely to be able to continue with most activities. Moderate depression requires four or more symptoms with the patient having great difficulty to continue with ordinary activities. Severe depression requires “several symptoms that are marked and distressing, typically loss of self-esteem and ideas of worthlessness or guilt. Suicidal thoughts and acts are common and a number of ‘somatic’ symptoms are usually present”.

As with the definition of the DSM-5 severity specifier, little research has been done on the ICD-10 severity specifier, perhaps because the reliability of making the severity distinctions is poor⁵⁶. Poor reliability is not surprising, due to the impreciseness of the severity level definitions⁵⁷.

The severity definitions in the official diagnostic systems have not been used in treatment studies. Rather, in almost all those studies, severity is designated by a score on a symptom rating instrument – usually the Hamilton Depression Rating Scale (HAMD)⁵⁸ or the Montgomery-Åsberg Depression Rating Scale (MADRS)⁵⁹. Thus, treatment studies generally do not consider other factors that have been used to characterize severity, such as level of functional impairment, degree of suicidality, or depressive subtype (i.e., presence of melancholic features or psychotic symptoms)^{60, 61}.

Scales measuring the severity of depression

The severity of depression has been most frequently quantified on paper-and-pencil and clinician-administered rating scales. There is variability amongst the instruments in the time frame covered (the two most common time frames being the past one or two weeks), rating guidelines (most scales use Likert-type ratings based on symptom frequency, persistence or intensity), and item content.

Little research has examined which parameters provide the most valid indicator of depression severity. Is the severity of depression best conceptualized as the number of symptoms (i.e., present or absent), frequency of symptoms (e.g., every day vs. half the days vs. few days), persistence of symptoms (e.g., always present vs. often present vs.

sometimes present), or intensity of symptoms (e.g., severe vs. moderate vs. mild)? Williams et al⁶², in standardizing the scoring of the HAMD, created a grid scoring format to incorporate information regarding symptom frequency/persistence and intensity in the ratings. The only study to examine whether it is important to consider both intensity and frequency constructs found that symptom intensity was a better indicator of severity than symptom frequency⁶³. In developing the Patient-Reported Outcomes Measurement Information System (PROMIS) depression scale, Pilkonis et al⁶⁴ reviewed studies comparing alternative response options and concluded that frequency scaling outperformed intensity ratings, though these were not studies of depression ratings. Thus, the most valid rating format of depression severity scales is unsettled, and has been little studied.

Should the content of a severity scale be based on the diagnostic criteria for the disorder, include other symptoms of depression that are not components of the diagnostic criteria (e.g., low motivation), or include symptoms that are frequent in depressed patients but are defining features of other disorders (e.g., anxiety, irritability)? And by what standard should one judge whether one approach or scale is a more valid indicator of severity? Statistical approaches such as item response theory have been used to construct scales⁶⁵, ⁶⁶. While instruments derived from this approach may be psychometrically superior to measures based on the diagnostic criteria for MDD, such measures do not include symptoms that have long been considered to be core components of depression, such as appetite and sleep disturbances or suicidality. If a measure of severity is to be utilized for clinical purposes, and not just for administrative outcome measurement, it is important to include vegetative symptoms, as the presence of these symptoms affects medication selection⁶⁷, and to assess suicidality because of safety concerns.

While there are differences amongst the scales in how they were constructed, their intended purpose, item coverage, and rating guidelines, the one commonality is that the overall severity of depression is represented by the sum of the ratings of the individual items. For all but a few scales, all items on the scale are rated similarly and contribute equally to the total score. A notable exception is the HAMD⁵⁸, which includes some items rated 0 to 2, and some others rated 0 to 4. To be sure, measures differ in their emphasis on different content domains of depression⁶⁸. Some measures have been criticized as being multidimensional, because a unidimensional construct of depression severity is

better able to demonstrate treatment effects⁶⁹. However, all scales, even multidimensional measures which yield subscale scores, as well as instruments that were initially intended to screen for depression rather than being used as indicators of severity, derive a total score that has been used to denote the severity of depression.

The *score summation approach* is based on some assumptions that have not been empirically supported. Adding up item scores to yield a total score as an indicator of overall depression severity assumes that all symptoms are equal indicators of the severity of depression. However, the different symptoms of depression are not similarly correlated with clinicians' global ratings of severity⁴⁸. From the psychometric perspective, the rating options of individual items should convey valid information across the entire spectrum of severity⁷⁰. Thus, severely depressed patients should more frequently receive the highest rating of a symptom than a low or zero rating, whereas mildly depressed patients should more frequently receive ratings indicating mild severity than the highest rating of a symptom. Santor and Coyne⁷⁰, using item response theory data analytic techniques, demonstrated that some of the items of the HAMD do not meet these assumptions.

In fact, scales based on item frequency ratings are unlikely to meet these assumptions and therefore may not be good measures of severity. For example, the items on the 9-item Patient Health Questionnaire (PHQ-9) are rated on a four-point scale of symptom frequency during the past two weeks: (0=not at all, 1=several days; 2=more than half the days; 3=nearly every day)⁷¹. Patients with MDD would be expected to score a 3 for most of the symptoms that are present, because the definition of MDD requires symptom presence for at least two weeks. Because of the ceiling effect, a patient with MDD seen in primary care who continues to work would score similarly to a depressed patient who is hospitalized because of difficulties with self-care. While there are several studies of the PHQ-9 using an item response theory approach, these have been of heterogeneous non-depressed psychiatric, medical or community samples^{72, 73, 74, 75, 76, 77, 78}. We are unaware of any studies evaluating the performance of the PHQ-9 items in a sample of depressed patients presenting for treatment. We would predict that, in such a sample, some – perhaps many – items of the PHQ-9 would be highly skewed because of the aforementioned ceiling effect. No studies have examined the impact of different rating guidelines on the operating characteristics of items on a depression scale.

Implicit in the score summation approach is that low level ratings across many symptoms reflect equal severity to high ratings across a fewer number of symptoms. For example, someone who indicates that, in the past week, he/she has infrequently experienced low mood, insomnia, low self-esteem, guilt, reduced concentration, fatigue, psychomotor slowing, insomnia, reduced appetite, reduced concentration, impaired decision making, and reduced interest in usual activities would be considered at the same level of severity as someone who reports daily depressed mood, guilt, feelings of inferiority, and suicidal thoughts, but denies all somatic and vegetative symptoms of depression. Likewise, when item ratings are based on symptom intensity, a mild intensity rating of many symptoms is considered the same as a severe intensity rating of a more limited number of symptoms.

The score summation approach, in which all items are weighted equally, is not grounded in a specific overriding conceptualization of severity. If illness severity is conceptualized in terms of mortality risk, then one would expect a measure of depression severity to weight more heavily item ratings of suicidal thoughts, hopelessness and psychomotor agitation than ratings of impaired concentration and fatigue. On the other hand, if illness severity is conceptualized in terms of functional impairment, then one might expect items assessing impaired concentration and fatigue to be weighted more heavily than items assessing appetite reduction or guilt. To be sure, some measures assess functional impairment along with symptomatology^{63, 71, 79, 80, 81}. No symptom-based measure, however, has been constructed by examining the association of individual items with indices of functional impairment and including on the scale only those items that are independently associated with impairment.

Few studies have examined the association between severity ratings of individual symptoms of depression and multiple external indicators of severity. Faravelli et al⁴⁸ found marked differences among symptoms in their association with CGI and GAF ratings. Moreover, the symptoms with the highest correlations with CGI ratings – such as depressed mood, psychic retardation, impaired concentration, and anhedonia – tended to have the highest correlations with GAF scores.

Most discussions of the problems with depression scales have focused on their limitations as outcome measures^{69, 82, 83}. However, different aspects of outcome measurement may be of interest, and these differences might result in different approaches towards scale

construction. Some measures of the severity of depression have been specifically designed to be sensitive to treatment effects^{59, 84}. Some measures are linked to the symptom criteria that are used to diagnose depression^{71, 79, 85, 86}, whereas others assess a broad range of features that patients indicate are most important in measuring outcome⁸⁰ or assess a range of diagnostic and associated symptoms of depression⁸⁷. Descriptions of scale construction typically focus on the content of the measure and rarely discuss the reason for choosing the rating format. For example, in developing the Multidimensional Depression Assessment Scale, Cheung and Power⁶⁸ reviewed the content of fifteen depression scales and how their scale would address a content gap. There was no discussion, however, of rating formats and why a symptom frequency format was chosen for their measure rather than a rating format assessing symptom intensity.

One of the commonly used clinician rated measures of severity, the MADRS, was designed to be particularly sensitive to change in treatment trials⁵⁹. Items were selected if they were prevalent in the patients at the beginning of treatment (i.e., prevalence greater than 70%), showed the greatest change from baseline to week 4 of treatment, and change in scores from baseline to week 4 on the symptom showed the greatest correlation with change in total scores on the measure. While there is nothing inherently wrong with constructing a measure in this manner for this purpose, this should not be the basis for selecting items on a measure of depression severity, as the resulting scale can be biased towards the inclusion of items that are particularly sensitive to change for the medication(s) studied. The construction of the MADRS was based on response to mianserin, maprotiline, amitriptyline, and clomipramine – medications that are not commonly used today. Using the same approach to construct a measure today, when different medications are prescribed, might produce a scale that only partially overlaps with the items included on the MADRS. In the same vein, the HAMD, which was published more than 50 years ago, has been criticized for including items that are most responsive to the effects of sedating medications such as tricyclic antidepressants⁸⁸.

So, while there are many rating scales of depression, and several studies examining them, questions remain as to how to judge if one measure is a more valid indicator of depression severity than another measure. Should it be based on psychometric analyses indicating unidimensionality? Would a “better” measure of severity be more highly correlated with indices of impairment? Be more highly correlated with current suicidal ideation? Be more

highly predictive of future suicidal behavior? Be more highly predictive of future mortality in general? Be more highly predictive of future course? Be better able to distinguish depressed patients who do and do not require hospitalization? Demonstrate a larger effect size in a treatment study? Have greater discriminative ability between depression and anxiety, and thus be a “purer” measure of depression?

A problem with depression scales: uncertain validity of cutoffs to define severity groupings

Putting aside the question of how to best conceptualize severity and construct a scale, a problem with the existing literature on depression severity is the inconsistency in the cutoff scores on symptom scales used to demarcate levels of severity, particularly severe depression. The use of various cutoff scores to define severity groups makes it difficult to compare the studies on the treatment implications of severity.

DeRubeis et al⁸⁹ conducted a mega-analysis of four studies comparing cognitive-behavioral therapy and medication, and defined severe depression as a cutoff of 20 or more on the 17-item HAMD. Likewise, the recent mega-analysis of placebo-controlled trials of fluoxetine and venlafaxine used a cutoff of 20 to define severe depression⁹⁰. Both of these studies cited the landmark study by Elkin et al⁹¹ to justify their definition of severe depression. However, Elkin et al did not cite empirical evidence for this cutoff and, in fact, did not refer to the patients scoring above 20 on the HAMD in absolute terms (i.e., having severe depression), but instead referred to these patients in relative terms (i.e., having more severe depression than the patients scoring 20 and below).

In Kirsch et al's⁹² meta-analysis of the impact of severity on antidepressant-placebo differences, the authors noted that the mean baseline HAMD scores of the antidepressant efficacy trials were in the very severe range (i.e., > 23) based on the American Psychiatric Association (APA)'s Handbook of Psychiatric Measures⁹³ for all but two of the 35 studies included in the analysis. In a prior analysis of antidepressant efficacy studies in the Food and Drug Administration (FDA) data base, Khan et al⁹⁴ divided the studies into three groups based on pre-treatment HAMD scores (<24, 25-27, >28) without indicating the basis for using these cutoff scores to define the groups. Fournier et al⁹⁵ used the thresholds recommended in the APA's Handbook of Psychiatric Measures⁹³ to define

grades of severity on the HAMD (mild to moderate: <18; severe: 19 to 22; very severe: >23). In contrast to these studies, and the APA guidelines, most pharmacotherapy studies have used a cutoff of 25 on the 17-item HAMD to define severe depression^{96, 97, 98, 99, 100, 101} and this cutoff has been recommended by several experts^{102, 103, 104}. Thus, severe depression has not been consistently defined.

Fundamental to studies on the treatment implications of severity levels is the validity of the cutoffs on the HAMD to define the severity categories. In none of the discussion sections of the meta-analyses and pooled analyses of the reports on severity and treatment outcome were questions raised about the cutoffs used to define the grades of severity. The APA's Handbook of Psychiatric Rating Scales⁹³ cited only two small studies in support of the cutoff scores to identify severity subtypes, and neither study provided support for the APA guidelines. One was a study examining the validity of deriving a HAMD equivalent score on the Schedule for Affective Disorders and Schizophrenia¹⁰⁵. This study did not attempt to determine the cutoff scores on the HAMD indicating grades of severity. The second study examined the association between HAMD scores and global ratings of severity in 59 depressed inpatients¹⁰⁶. The authors did not derive (or recommend) cutoff scores corresponding to severity levels. Thus, it is unclear why a cutoff of 19 was recommended in the APA Handbook to identify severe depression. The UK National Institute for Health and Clinical Excellence (NICE) guidelines recommended a cutoff of 23 to identify severe depression on the HAMD, though no research was cited to support this recommendation¹⁰⁷.

Because of the limited amount of empirical research establishing cutoff scores for bands of severity on the HAMD, and the significance accorded to severity by treatment guidelines, our clinical research group also examined this issue in 627 psychiatric outpatients with MDD who were rated on the CGI¹⁰⁸. The cutoff score on the HAMD that maximized the sum of sensitivity and specificity was 17 for the comparison of mild vs. moderate depression and 24 for the comparison of moderate vs. severe depression. Based on a review of the available evidence, as well as the recommendations that a cutoff of 7 be used to define remission, we recommended the following severity ranges for the 17-item HAMD: no depression (0-7); mild depression (8-16); moderate depression (17-23); and severe depression (>24).

Each of the above studies derived cutoff scores based on clinicians' global judgments of severity. A limitation of these studies is that it is not known on what basis the global judgments of severity were made. Were some symptoms of depression considered better indicators of severity than other symptoms? For example, are symptoms characteristic of melancholic or endogenous depression given greater weight in clinicians' CGI ratings? Are clinicians' global ratings disproportionately influenced by degree of suicidality? Do clinicians consider psychosocial impairment in making their CGI ratings? We are unaware of any studies that have attempted to derive severity ranges on the HAMD, or any other depression scale for that matter, based on degree of impairment or level of suicidality.

Another problem with depression symptom scales: different scales classify patients into different severity groups

In clinical practice, self-report questionnaires are preferable to clinician-rated scales because they take less time to administer. If self-report scales are to be used to classify patients into severity categories, and if treatment recommendations are to be based, in part, on severity classification, then it is important for different scales to classify individuals similarly. However, because the content of measures differ, it would not be surprising if there were significant differences between measures.

Cameron et al¹⁰⁹ compared the PHQ-9 and the Hospital Anxiety and Depression Scale (HADS) severity classifications in a sample of primary care patients referred by their general practitioners in the UK to a mental health worker¹¹⁰. No information was provided regarding the patients' psychiatric diagnoses. They found that the PHQ-9 overclassified severity compared to the HADS, with twice as many patients classified in the severe range. Other studies comparing the PHQ-9 and the HADS in medical patients found similar results^{111, 112}. However, these studies lack an external validator and it is therefore unclear if the PHQ-9 overclassifies, or the HADS underclassifies, severity. A second study by Cameron et al¹⁰⁷ included the second edition of the Beck Depression Inventory (BDI-II)¹¹³ along with the PHQ-9 and HADS, and also assessed the patients with the HAMD. The participants were primary care patients who had been diagnosed by their general practitioner with depression. Both the PHQ-9 and BDI-II overclassified severity compared to the HAMD, whereas the HADS underclassified severity.

We are aware of only one study that compared self-report scales in a sample of psychiatric outpatients with MDD¹¹⁴. Our clinical research group compared severity classification on three measures that assess the DSM-IV/DSM-5 symptom criteria for MDD: the Clinically Useful Depression Outcome Scale (CUDOS)⁷⁹, the Quick Inventory of Depressive Symptomatology (QIDS)⁸⁵, and the PHQ-9⁷¹. The patients were also rated on the 17-item HAMD. In a study of depressed outpatients, we found that the correlations between the HAMD and all three self-report scale scores were nearly identical, and the average correlation among the three self-report scales was .73. However, the scales significantly differed in their distribution of patients into severity categories. Approximately one-third of the patients scored in the mild range on the HAMD and CUDOS, whereas approximately 10% of the patients were mildly depressed according to the PHQ-9 and QIDS. On the CUDOS and HAMD, moderate depression was the most frequent severity category, whereas on the PHQ-9 and QIDS the majority of the patients were classified as severe. The majority of the patients in the moderate range on the HAMD were in the severe range on the PHQ-9 and QIDS. Significantly fewer patients were classified as severely depressed on the CUDOS compared to the PHQ-9 and QIDS.

With the three self-report measures being highly correlated with each other, and equally correlated with the HAMD, what, then, might account for the marked differences between scales of similar content in the distribution of patients into severity groups?

The cutoffs on the three scales to define the severity groups were derived in different ways, and this was likely responsible for the differences between the scales in severity classification. For example, Kroenke et al⁷¹ indicated that the cutoff scores on the PHQ-9 were chosen for the pragmatic reason of making them easier for clinicians to recall. They also noted that alternative cutoffs did not increase the association between increasing PHQ-9 severity and indices of construct validity. When selecting the cutoff scores to define the severity ranges on the PHQ-9, the developers of this questionnaire did not consider the potential impact of the broadness by which severity ranges were defined and how this might impact on treatment recommendations of official treatment guidelines.

Kroenke et al⁷¹ indicated that, when severity groupings based on different cutoffs are equally associated with external variables, then the cutoffs can be chosen based on their ease of recall. We disagree with this reasoning. For all scales measuring the severity of

depressive symptoms, the thresholds distinguishing patients with mild, moderate and severe depression do not represent well-demarcated lines separating the severity subtypes. As with other areas of psychopathology, the severity of depression better corresponds to a dimensional than a categorical model of classification¹¹⁵. Thus, alternative cutoffs to categorize severity groupings are likely to also be valid when the groupings are compared on an external variable such as psychosocial functioning. However, one should not be cavalier about the choice of cutoffs, because they impact on the relative broadness of each of the severity categories.

If clinicians are to follow official treatment guidelines' recommendations and base initial treatment selection on the severity of depression, then it is important to have a consistent method of determining depression severity. The marked disparity between standardized self-administered scales in the classification of depressed outpatients into severity groups indicates that there is a problem with the use of such instruments to classify depression severity. If official treatment guideline recommendations were followed, then use of measures such as the QIDS and PHQ-9, which broadly define the severe category, would result in greater reliance on medication in preference to psychotherapy as the first line treatment option for MDD. Caution is thus warranted in the use of these scales to guide treatment selection until the thresholds to define severity ranges have been better established empirically.

The importance of severity of depression in treatment: official guideline recommendations

Notwithstanding the aforementioned problems with conceptualizing the severity of depression, and defining the cutoffs on scales for severity levels, depression severity is an important consideration in treatment decision-making. The severity of depression has influenced treatment recommendations in official guidelines. The third edition of the APA's guidelines for the treatment of MDD recommend both psychotherapy and pharmacotherapy as monotherapies for depression of mild and moderate severity, and pharmacotherapy (with or without psychotherapy) for severe depression¹. The NICE updated guidelines for the treatment and management of depression discourage the use of antidepressant medication as the initial treatment option for mild depression, and recommend medication together with empirically supported psychotherapy for moderate

and severe depression². As reported by van der Lem et al¹¹⁶, treatment guidelines in the Netherlands also recommend pharmacotherapy as the first treatment option for severely depressed patients, and either pharmacotherapy or psychotherapy for mildly and moderately depressed patients. While the recommendations in these guidelines are not entirely consistent, they are unanimous in recommending medication as the treatment of choice for severe depression.

The treatment significance of severity has been studied in several different ways. There are controlled studies, effectiveness studies, pooled analyses, and meta-analyses examining the impact of severity on particular treatments^{117, 118, 119, 120, 121, 122}, comparing treatments across a range of severity^{99, 123, 124, 125, 126, 127}, comparing medication and placebo across a range of severity^{128, 129}, comparing psychotherapy and control groups across a range of severity^{130, 131}, comparing treatments amongst severely depressed patients^{96, 101, 102, 132}, and examining whether severity predicts short-term outcome^{42, 133, 134, 135}, treatment resistance¹³⁶, longer-term outcome^{40, 137, 138, 139}, and relapse³⁸.

Severity of depression and pharmacotherapy

In the past decade, questions have been raised whether selective serotonin reuptake inhibitors (SSRIs) and other new generation antidepressants are effective in non-severe depression. Khan et al⁹⁴ analyzed 45 clinical trials in the FDA database and found that in studies with a mean baseline 17-item HAMD score of 24 or less there was little evidence that antidepressant medication was superior to placebo, whereas in studies with a mean baseline HAMD score of 28 or greater there was clear evidence that medication was superior to placebo. Kirsch et al⁹² similarly examined the FDA database, and they also examined the efficacy of antidepressants as a function of mean baseline HAMD score in the trial. Their results largely replicated the findings of Khan et al⁹⁴ that drug-placebo differences were largest in the studies with the highest baseline severity (i.e., HAMD >28). Kirsch et al⁹² found that antidepressants were significantly more effective than placebo in the less severe cohorts, but they considered the difference in response to be modest and clinically insignificant.

In contrast to the analyses of the FDA database by Kirsch et al⁹² and Khan et al⁹⁴,

Fournier et al⁹⁵ pooled individual patient data from six published studies. Kirsch et al and Khan et al used aggregated mean scores for an entire study as the unit of analysis. That is, they compared studies with different mean severity scores at baseline. The problem with this approach is that a group of patients with a mean score in the severe range will also include some patients in the mild and moderate severity ranges. Likewise, a group of patients with a mean score in the mild or moderate severity range will include some patients scoring in the severe range. Pooling individual patient data avoids the problem of severity group misclassification at the individual patient level. Fournier et al⁹⁵ replicated the finding that drug-placebo differences were clinically significant only for severely depressed patients, and found only a small effect size for mildly and moderately depressed patients.

More recently, other pooled analyses of patient level data (rather than aggregated data from a trial) have been conducted. Using pharmaceutical company data bases, these analyses included all studies of a product, thereby avoiding the bias inherent in examining only published studies¹⁴⁰. The results of three large, pooled analyses of published and unpublished studies, which included between 4,000 and 10,000 subjects each, indicated that antidepressants are effective across a range of severity^{90, 129, 141}. These analyses, and the controversy that has been stirred regarding the efficacy of antidepressants, highlights the impact that considerations of severity might have on clinical practice.

Severity of depression and medication or psychotherapy as first line treatment

A second important severity related treatment question is whether the severity of depression should be used as the basis for recommending medication or psychotherapy as first line treatment. More specifically, the question is whether patients with severe depression should preferentially be treated with medication. A related question is whether psychotherapy is beneficial for severely depressed patients.

Symptom severity as a moderator of treatment response has been the subject of ongoing debate since the publication of the results from the US National Institute of Mental Health Treatment of Depression Collaborative Research Program (TDCRP), suggesting that psychotherapy was not as effective as medication in the acute treatment of severe

depression^{91, 142}. The first meta-analysis of studies directly comparing psychotherapy and pharmacological interventions included 30 published studies of more than 3,000 patients¹⁴³. A meta-regression analysis examining whether effect sizes were associated with mean baseline scores on the HAMD or BDI found no evidence that baseline severity was associated with differential treatment outcome. A comparison of effect sizes in studies with baseline HAMD scores below 20 vs. 20 and above also found no differences.

A meta-analysis of 132 controlled psychotherapy studies of more than 10,000 patients found that greater mean baseline symptom severity did not predict poorer response¹³⁰. More recently, Weitz et al¹⁴⁴ pooled individual patient data from 16 studies comparing antidepressants and cognitive behavior therapy. They defined the severe group according to the APA (HAMD ≥ 19) and NICE (HAMD > 23) recommendations. Increased severity was associated with significantly lower remission rates (but not response rates) in both the medication and psychotherapy treatment conditions. Severity was not associated with differential treatment outcome, thus confirming the results of a prior pooled analysis based on a smaller number of studies⁸⁹. In a follow-up study, the authors conducted a pooled analysis focused on the five studies that used placebo as the control condition¹³¹. The results were consistent with the larger pooled analysis: baseline symptom severity was not associated with change in symptom severity scores from baseline to endpoint between the cognitive behavior therapy and pill placebo groups.

The results of these more recent meta-analyses, based on severity classification according to symptom rating scales, are thus not consistent with official treatment guidelines which recommend medication as the first line treatment for severe depression.

SEVERITY OF PERSONALITY DISORDERS

Severity is clearly of import to PDs, though the current diagnostic systems do not include any formal severity ratings. PD patients identified as “severe” are more likely to exhibit high comorbidity with other psychiatric diagnoses, particularly mood, anxiety, substance use¹⁴⁵, and other PDs¹⁴⁶. So-called “severe” cases are often in treatment for protracted periods of time^{147, 148, 149}, exhibit higher rates of hospitalization and suicide attempts¹⁵⁰, and self-injure with greater frequency¹⁵¹. They are likely to be incarcerated, unable to hold down a job, and have failed relationships¹⁵². It is generally agreed that

they may present a public health burden, and therefore should be identified early and get treated often^{3, 4, 153}.

Nonetheless, the question remains: what is meant by “severe” PD? Severity has been assessed by counting the number of comorbid PD diagnoses overall, with higher comorbidity indicating higher severity^{152, 154, 155, 156}. However, this may better reflect the severity of overall personality pathology rather than the severity of a particular PD. More severe cases of personality pathology may further be identified by case complexity and specific comorbidity patterns. The main section of DSM-5 (i.e., Section II) identifies PDs as occurring in one of three clusters. Tyrer and Johnson¹⁵⁷ proposed that individuals with comorbid PDs from more than one cluster are more severe than those with comorbid PDs from the same cluster. The authors further identify antisocial PD as the most severe PD based on risk to others. Therefore, the most severe cases must be diagnosed with antisocial PD as well as PDs from other clusters. Using this model, severity of PD was associated with conduct disorder, criminal behavior, homelessness, institutionalization, unemployment, and delinquent behavior in childhood.

Severity of a specific PD may be measured by counting the number of criteria met. For example, cases of borderline PD for which nine criteria are endorsed would be viewed as more severe than patients endorsing only five criteria¹⁴⁷. However, results from our clinical research group did not support this hypothesis, finding no differences in comorbidity or psychosocial functioning based on criteria count for patients diagnosed with borderline PD¹⁵⁸. Alternatively, severity can be defined by the frequency of symptoms. For instance, patients with borderline PD who engage in self-injury multiple times daily would be more severe than those reporting only monthly self-injury¹⁵¹.

Specific PDs have even been identified as more or less severe than others. Kernberg and Caligor¹⁵⁹ organized PDs into a hierarchy ranging from “more severe” (e.g., borderline PD) to less severe (e.g., obsessive compulsive PD, dependent PD). There has also been a strong push for conceptualizing PDs using constellations of pathology personality traits. From this perspective, a “severe” PD symptom or trait may be defined as one that is statistically extreme, or existing in only a very small proportion of the population¹⁶⁰.

Treatment research of “severe” personality disorders primarily emphasizes symptom

characteristics (frequency, persistence, intensity) and functional impairment (social/occupational, or outcomes such as imprisonment)[161](#), [162](#), [163](#). Maden and Tyrer[162](#) identify a category of “dangerous and severe” PD, which is characterized by having a high risk of causing unrecoverable harm to others. Confusingly, the first criterion for having a “dangerous and severe” PD is already being diagnosed with a “severe disorder of personality” which remains undefined itself. The authors do not clarify what severity means at the criterion level, although it appears this definition is legal in origin, and refers primarily to psychopathy and not to PDs as they are traditionally defined.

Severity of personality disorders and functioning

Although severity has been defined in various ways in the PD literature, a general consensus appears to have emerged that PD severity is inherently linked with level of maladaptive functioning[164](#), [165](#), [166](#), [167](#), [168](#), [169](#). It is widely acknowledged that extreme trait or symptom variation is insufficient to diagnose PDs or to dictate diagnostic severity. Rather, the emphasis lies in having extreme personality traits in the presence of impairment associated with those traits. Unlike physical illnesses, or even depression, which are more focused on symptom presentation, personality diagnoses are intertwined with adaptive functioning. Like depression, PDs by definition must result in “distress or impairment” to be diagnosed[33](#). In contrast to depression, however, the symptom criteria for diagnosing PDs include both affective/cognitive/emotional *and* functional components. For example, impoverished occupational and financial functioning is included in symptom criteria for antisocial PD, and failure to engage in social and leisure activities is part of the criteria for obsessive-compulsive PD.

The interrelationship between functional impairment and personality leads many to conclude that PD severity is a combination of extreme personality disturbance and maladaptive functioning associated with that disturbance[165](#), [169](#). In fact, functioning is so fundamental to determining PD presence and severity that some authors argue that assessing extreme traits/symptoms is unnecessary[170](#), [171](#), [172](#), [173](#). Thus, one need not demonstrate symptom severity if sufficient impairment is judged to be present. However, the dysfunction must be determined as due to the presence of the personality features, even if they are not extreme. For example, using the multi-axial DSM-IV, Livesley[174](#) proposed defining PD as present diagnostically on Axis I, and coding personality traits

separately on Axis II. Widiger and Trull¹⁶⁹ proposed a similar model, only using the GAF score on Axis V as a stand in for severity.

Taken together, these models converge on defining severity as a generalized, adaptive failure of an intrapsychic system required to fulfill daily life tasks¹⁶⁶. Although specific areas of impairment differ, there is convergence on impairment in three broad areas: identity formation, self-control (or direction), and interpersonal relationships¹⁶⁴. However, some research indicates that pathological personality traits and functioning are so closely intertwined that they may not represent distinct domains¹⁷⁵.

Severity of personality disorders as described in DSM-5 and ICD-10

There is no clear mention of severity with respect to PDs in the main section II of DSM-5³³. However, the overall description of PDs includes severity indicators common to other disorders. For example, PDs are specifically noted to be inflexible, maladaptive, pervasive, and associated with “clinically significant” functional impairment or subjective distress. Functional impairment is an indicator of severity in many physical and psychiatric disorders; pervasiveness is a severity indicator for depression; and subjective distress is identified as indicating a “severe case” for disorders of mood and sexual function. As it stands, there is no official method for indicating PD severity in DSM-5.

Section III (Emerging Measures and Models) of DSM-5 includes an alternative model for diagnosing PDs. Diagnosis is defined via a combination of severity levels of dysfunction and elevated personality traits, and severity is determined principally by dysfunction associated with elevated traits³³. This model does not designate a measure for overall severity, but “moderate or greater impairment” is required for diagnosis. Impairment is operationalized as falling into one of five levels, with the extreme end indicative of severe personality pathology. The Level of Personality Functioning Scale (LPFS) is proposed to rate impairments in functioning, and therefore also PD severity. Ratings are made for self (identity and self-direction) and interpersonal (empathy and intimacy) functioning. Levels include: 0 (little or no impairment), 1 (some impairment), 2 (moderate impairment), 3 (severe impairment), 4 (extreme impairment). Individuals with extreme impairment are described as having an impoverished, unclear identity and self-direction with maladaptive self-concept, and completely lacking capacity to engage interpersonally.

Interestingly, DSM-5 Section III also includes discussion of an additional measure of personality traits, the Personality Inventory for DSM-5¹⁷⁶. The items are clearly trait content related; however, the measure provides an overall summed score identified as measuring “overall personality dysfunction”. The identification of extreme traits as indicative of dysfunction is curious, but not inconsistent with the significant overlap between functioning and PD traits/symptoms found elsewhere in the literature¹⁷⁵. Nonetheless, this suggests that extreme traits are at least indicative of extreme dysfunction, which is the primary index of severity in this model.

Similar to the DSM-5, the ICD-10 does not make mention of severity in PD classifications. However, several papers have been published on changes proposed for ICD-11, which are substantial. Most notably, the primary classification of PDs will change to one based on severity of personality disturbance. Description of PD traits or features is optional but will not be required for diagnosis^{3, 4}.

Consistent with the larger literature, the proposed changes to the ICD-11 conceptualize severity primarily as dysfunction, or the personality-related problems experienced by the individual. Again, five levels of severity are proposed, though they vary slightly from those in the DSM. Summed together, severity levels are dictated first by pervasiveness of the impairment (across situations or limited), and secondarily by the number of problematic personality traits (multiple or single). At the highest level of severity, risk to self or others is also assessed. Thus, the most severe cases are identified by functioning above all else. Symptoms/traits and risk of harm are secondary, but also considered. Unlike the DSM-5 alternative model proposal, dysfunction in self and identity is not included in severity ratings^{3, 4}. At the time of this writing, the ICD-11 has not yet been published, and therefore these definitions should be considered provisional. Nonetheless, the emphasis on functioning via severity ratings has been criticized for insufficient research establishing its reliability and validity¹⁷⁷.

Measures of personality severity

As early as 1996, Tyrer and Johnson¹⁵⁷ developed a five-point scale assessing disorder severity similar to that in the ICD-11 proposal. Ratings were made based on information derived from a trait personality measure, the Personality Assessment Schedule (PAS)¹⁵³.

Thus, severity was weighted more towards extremity on traits than on functioning. The PAS has also been used to classify individuals into the four PD categories proposed by Tyrer and Johnson¹⁵⁷: no PD, personality difficulty, simple PD, complex PD. PAS severity designations are primarily based on the frequency of DSM-IV and ICD-10 categories, and have been used in studies predicting treatment outcomes, albeit with mixed findings¹⁷⁸. The General Assessment of Personality Disorder¹⁷⁹ has been used as an index of severity in multiple studies, and provides two main scales of severity – self-pathology and interpersonal problems – both of which reflect functional impairment as defined by the DSM-5^{164, 180, 181}. Similarly, the Severity Indices of Personality Problems¹⁷³ defines severity as a combination of impoverished self and interpersonal functioning.

Relatively few measures of severity exist for individual PDs, and these largely focus on borderline PD. For example, the Borderline Personality Disorder Severity Index (BPDSI)^{151, 182} is a semi-structured clinical interview that operationalizes severity primarily by frequency of borderline PD symptom behaviors over the preceding three months. Frequency of symptoms is rated from 0 (never) to 10 (daily). Severity is averaged across these scores, yielding severity scores for individual borderline PD criteria as well as the diagnosis overall. Thus, the BPDSI largely measures severity as a function of symptom frequency, though many of the items also ask about behaviors that have implied functional consequences (e.g., going out instead of working).

Consistent with the severity of personality pathology often being linked with impairments in functioning, PD treatment outcome research has often focused on the degree to which various treatment approaches (e.g., dialectical behavioral therapy, mentalization-based treatments, transference-focused psychotherapy) improve day-to-day functioning and reduce specific, concrete maladaptive behaviors^{147, 183, 184}. For instance, in the extensive borderline PD treatment literature, change in personality pathology is often assessed using measures such as the Zanarini Rating Scale for Borderline Personality Disorder¹⁸⁵ and the Barratt Impulsiveness Scale¹⁸⁶. However, reduction in suicide attempts, self-harm behavior, and reliance on psychiatric emergency treatment services are often primary treatment outcome measures, as are improvements in maintaining meaningful relationships and improving workplace functioning^{147, 183, 184, 187, 188}.

Although the PD treatment literature has focused primarily on the treatment of borderline

PD, other PDs also have received some attention, with functional impairment being identified as central to treatment outcomes. For instance, transference-focused psychotherapy has demonstrated some benefit for patients with comorbid narcissistic and borderline PD, and this treatment approach emphasizes interpersonal functioning in personal and workplace relationships when assessing outcome¹⁸⁹. Treatment research on antisocial PD has focused on subsequent substance use and arrests¹⁹⁰. Thus, across the treatment of various PDs, treatment outcome and a reduction in “severity” is understood not just as symptom reduction, but also reduction in specific deleterious behaviors (e.g., self-harm) and the promotion of interpersonal functioning and specific prosocial behaviors (e.g., maintaining employment).

TRANSDIAGNOSTIC MODELS AND SEVERITY: THE EMERGENCE OF PSYCHOPATHOLOGY SPECTRA

Many of the questions asked above as to how to compare the validity of depression scales in measuring severity also apply to determining if different diagnoses confer differential levels of severity. The likelihood of meeting criteria for different diagnoses confers standing on underlying genetic liabilities¹⁹¹, ¹⁹². This is important to consider given that individuals who meet criteria for one diagnosis are very likely to meet criteria for multiple other diagnoses¹⁹³, such that various diagnoses may be thought to be manifestations of underlying spectra (e.g., antisocial PD, narcissistic PD and substance use all reflect an underlying externalizing spectrum).

Research examining the relations amongst various internalizing diagnoses characterized by subjective distress and fear suggests that it may be “easier” for individuals to meet criteria for diagnoses such as MDD than for more “severe” disorders such as generalized anxiety and panic disorders¹⁹⁴. Put differently, meeting criteria for generalized anxiety or panic disorder reflect higher standing on the internalizing dimension than would simply meeting criteria for MDD. Interestingly, Krueger and Finger¹⁹⁴ also found that high standing on the internalizing dimension was linked robustly to lifetime number of inpatient hospitalizations and past month psychosocial functioning.

Other more recent research has also linked “severity” on the internalizing spectrum to key outcomes. For instance, Eaton et al¹⁹⁵ found that the likelihood of meeting criteria for

various depressive disorders, anxiety disorders, and bipolar disorders can be represented by an underlying continuum. Individuals with high scores on this dimension, who would be characterized as having more “severe” levels of internalizing psychopathology, would thus be likely to meet criteria for many diagnoses and to report a broad range of symptoms (e.g., depressed mood, worry, concentration difficulties, irritability) characterizing the various DSM diagnoses defining this dimension.

Eaton et al¹⁹⁵ presented evidence indicating that scores on the internalizing spectrum predicted outcomes such as the future occurrence of internalizing symptoms (e.g., depressed mood, worry), suicide attempts, angina/chest pain, and ulcers. Moreover, standing on this underlying dimensionally-based internalizing spectrum predicted these outcomes much more strongly than did DSM-based conceptualizations of various internalizing disorders (e.g., MDD, generalized anxiety disorder), thereby providing evidence for the utility of this approach in capturing severity as it relates to important outcomes such as suicidality and physical health concerns¹⁹⁵.

In regard to other forms of psychopathology, Krueger et al¹⁹⁶ presented evidence indicating that symptoms and behaviors defining personality and substance use disorders can be captured by an underlying externalizing dimension. Other research also supports the presence of this underlying latent externalizing dimension, which explains why antisocial behaviors (e.g., various unlawful behaviors) and traits (e.g., impulsivity, callousness) and substance use issues are likely to co-occur^{191, 197}. Carragher et al¹⁹⁷ presented findings suggesting that meeting criteria for some disorders (e.g., cocaine dependence) confers higher standing and severity on this underlying externalizing dimension than other “less severe” disorders (e.g., nicotine and alcohol dependence). Similarly, overlap in disorders such as schizophrenia and schizotypal PD appears to be reflected by a thought disorder spectrum^{191, 198}. Standing on this spectrum has been linked to functional impairment and illness course¹⁹⁸.

Going forward, it will continue to be important for future research to further explicate how level of severity (i.e., how likely an individual is to meet criteria for different disorders and to meet criteria for “difficult” disorders such as cocaine dependence in the case of the externalizing spectrum) captured by broad internalizing, externalizing, and thought disorder dimensions predicts illness course and other key outcomes related to

morbidity and mortality. These dimensions account for diagnostic comorbidity amongst various disorders and have been shown to predict various outcomes more strongly than diagnostic status on various DSM disorders, suggesting important merits to this approach^{191, 195}. In this regard, the Hierarchical Taxonomy of Psychopathology (HiTOP) has emerged as a dimensionally-based alternative to the DSM-5^{191, 199}. Thus, it will be important to determine the degree to which this framework adequately captures psychopathology “severity”, however severity is defined, and is useful for researchers and practitioners.

CONCLUSIONS

The issue of severity has great clinical importance. Severity influences decisions about level of care and affects decisions to seek government assistance due to psychiatric disability. In outpatient settings, the importance of severity is reflected in the controversy about the efficacy of antidepressants across the spectrum of depression severity, and whether patients with severe depression should be preferentially treated with medication rather than psychotherapy.

We began this paper with a series of questions as to how the severity of psychopathology should be conceptualized. Some authors have suggested that the core indicator of the severity of mental illness is functional disability²⁰⁰. The DSM-5 has defined the severity of different disorders in different ways. Our review of the literature for depression and PDs demonstrated that researchers have adopted a myriad of ways of defining severity. The severity of depression has predominantly been defined according to scores on symptom rating scales. To be sure, there is some variability in how items are rated (i.e., symptom intensity vs. symptom frequency vs. symptom persistence), as well as some variability in the range of symptoms assessed by different measures of depression. Irrespective of the precise manner by which symptom severity is determined, most of the literature on the severity of depression is based on the parameters of symptoms. By contrast, the core of personality pathology is intertwined with its impact on functioning. Distinguishing extreme variants of personality traits from functioning has been challenging, therefore functional impairment has been fundamental to conceptualizing the severity of PDs.

Because the functional impact of symptom-defined disorders such as MDD depends on factors unrelated to the disorder such as self-efficacy, resilience, coping ability, social support, cultural and social expectations, as well as the responsibilities related to one's primary role function and the availability of others to assume those responsibilities, we would argue that the severity of such disorders should be defined independently from functional impairment. To those who would disagree, consider the following scenario: two individuals have an upper respiratory tract infection. They have the same elevation in body temperature, sneeze and cough with the same frequency, have the same level of mucus production and nasal discharge, and the same viral load. And the symptoms last for the same number of days. In short, they have the same intensity, frequency, and persistence of symptoms. Yet one person misses work for a week and the other does not miss work. Does the person who missed work have a more severe upper respiratory tract infection?

A distinction could be made between defining severity at the level of a disorder vs. overall global illness severity. As stated, at the level of disorder, severity should be determined by the factors that are intrinsic to the disorder. Thus, the severity of depression should be determined by the intensity, frequency, and/or persistence of the depressive symptoms. And the same is true for other disorders such as generalized anxiety disorder, post-traumatic stress disorder, mania/hypomania, and tic disorder. The severity of panic disorder should be based on the intensity and frequency of panic attacks. The severity of premature ejaculation should be based on time to ejaculation, the severity of hypoactive sexual desire based on the intensity (or lack thereof) of desire, the severity of binge eating disorder on the frequency and intensity of binges, etc.. The episodic nature of some psychiatric disorders and symptoms presents some measurement challenges. There may be day-to-day variability in symptom intensity as well as symptom persistence through the course of the day. Symptom frequency varies by disorder. Too little research has compared the validity of symptom intensity, frequency, and persistence assessments.

Severity, however, can be considered from another perspective: at the level of overall illness. A patient with depression, borderline PD, some anxiety disorders, substance use disorder and an eating disorder has a severe illness. It would likely be difficult to parse the levels of functional impairment to the separate disorders. The severity of the symptoms of depression may not be high, but the patient is nonetheless severely ill. How to take into

account comorbidity when determining the severity of individual disorders is not clear. A global rating of overall illness severity was included in DSM-III through DSM-IV, but dropped from DSM-5. The global rating of illness severity can be considered to be akin to the composite measures of physical illness severity, described in the introduction, that have been used to predict mortality in emergency room and hospitalized patients. The problem with the GAF was that it was a single rating that required consideration of multiple constructs, including symptom frequency, type of symptom, level of impairment, suicidality, ability to care for oneself, and psychosis. Because of its complexity, there were problems with the reliability of its ratings²⁰¹. Perhaps the dimensionally based measures of psychopathology articulated in HiTOP will yield clinically meaningful and useful approaches towards characterizing overall severity.

In the future, research on severity needs to be clear as to what correlates of a measure are expected. We noted above that too little research has compared the validity of symptom intensity, frequency, and persistence assessments. The question is how to evaluate validity. Should severity be a predictor of outcome? Should it help match patients to appropriate treatments or appropriate levels of care? Should it predict mortality? Should it reflect underlying pathophysiology? Should it confer genetic risk? Should it be used to guide the allocation of finite resources at either the insurance company or governmental funding agency level?

There are a wealth of papers in the psychiatric, medical and epidemiological literatures that refer to depression severity in the title and examine the correlates of a measure of depressive symptoms. But how to best measure severity has largely not been the subject of study. Numerous scales have been developed that purport to measure the severity of depression. When the authors of these scales discuss the reason behind developing their measure, the explanation usually focuses on item content and rarely discusses the reason for choosing a particular rating approach. Perhaps it does not make a meaningful difference how items are scaled. Perhaps the exact content of a scale does not make a meaningful difference either. Perhaps simplicity and clinical utility should trump any minor incremental validity that one measure shows over another.

However, some research suggests otherwise. The ability to detect differences between medication and placebo may be related to the content of the measure used²⁰². Scales

differ in severity classification^{111, 112, 114}, and treatment guidelines suggest that severity be used to select among treatment alternatives^{1, 2}. Thus, severity has real world implications in both the research and clinical communities. It is our hope that this paper stimulates more consideration and research into the issue of how to best conceptualize and measure the severity of psychiatric disorders.

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